AN EXTRAPOLATIVE DIAGONALIZATION OF INCOMPLETE HAMILTONIANS

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We propose a new technique of constructing the low-lying bound states. It starts from an "available" non-square finite submatrix of the hamiltonian and its essence lies in a preliminary band-matrix iterative extrapolation of these matrix elements. The efficiency of the method is illustrated on the standard example – the anharmonic oscillator.

In the nuclear, atomic, molecular or even relativistic quantum field physics [1], computer diagonalization of hamiltonians is limited by the available storage and by the time needed for an evaluation of the matrix elements. Thus, after the choice of a suitable basis, we always encounter the question of the relevance and physical importance of the separate matrix elements. Usually, it is solved by the simple square-matrix truncation based on the standard variational ansatz of the type

$$|\psi\rangle = \sum_{n=0}^{N} |n\rangle\,\langle n|\psi^{[N]}\rangle, \quad N\to\infty . \qquad (1)$$

A priori, the most relevant matrix elements of H need not necessarily form a square matrix. As an illustration, we may recall any perturbed hamiltonian $H=H_0+H_1$ with diagonal H_0 and small H_1. Rather surprisingly, the first-order perturbation prescriptions of the type

$$|\psi\rangle \approx |0\rangle + \sum_{m=1}^{M} |m\rangle \frac{\langle 0|H|m\rangle}{\langle 0|H_0|0\rangle - \langle m|H_0|m\rangle} \qquad (2)$$

contain the whole column of the hamiltonian matrix rather than a square submatrix. Similarly, the first-order degenerate extensions of (2) necessitate the oblong sets of matrix elements

$$\langle n|H|m\rangle, \quad m=0, 1, ..., M,$$
$$n=0, 1, ..., N\ (<M) . \qquad (3)$$

with $M\to\infty$ in principle. Of course, an $M\to\infty$ convergence in the formulas of type (2) implies also the existence of a reasonable cut-off $M<\infty$ specifying (3) as a main source of the relevant physical information, say, about the low-lying spectrum of the hamiltonian H. This will be our basic assumption here.

Using (3) finds independent support in the phenomenological success of the so-called chain models [2] and techniques [3] based on the so-called doorway state hypothesis

$$\langle n|H|m\rangle \approx 0, \quad m>M>N\geqslant n . \qquad (4)$$

It enables us to write the first $N+1$ rows of the rigorous Schrödinger equation as relations coupling just the first $M+1$ projections of the wavefunctions. For example, with the orthonormalized basis states $|n\rangle$, we have

$$\sum_{m=0}^{M} (\langle n|H|m\rangle - E\delta_{mn})\langle m|\psi\rangle = 0,$$
$$n=0, 1, ..., N . \qquad (5)$$

Now, our second assumption may be formulated as an expectation that, for some sufficiently large parameter $N\gg 1$, the functions $\langle n|H|m\rangle$ of indices m and n remain sufficiently smooth for $m>M$ and $n>N$.

Thus, we may formulate our main idea as a completion of eq. (5),

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$$\sum_{m=0}^{\infty} (\langle N+k|H|m\rangle - E\delta_{N+k,m})\langle m|\psi\rangle = 0,$$

$$k=1, 2, \ldots . \tag{6}$$

by means of an extrapolation evaluation of the necessary matrix elements of H from the "available" input (3).

Formally, the general idea of extrapolation is rather ambiguous. For the sake of definiteness, let us start from the simplest possible ansatz

$$\langle n|H|m\rangle \approx \langle n+1|H|m+1\rangle,$$

$$n \geqslant N, \quad m \geqslant M, \tag{7}$$

and notice that such an extrapolation prescription converts (6) into an ordinary difference equation

$$\langle N|H|0\rangle\psi_0 + \langle N|H|1\rangle\psi_1 + \ldots$$

$$+ \langle N|(H-E)|N\rangle\psi_N + \ldots$$

$$+ \langle N|H|M\rangle\psi_M = 0, \tag{8}$$

with $\psi_m = \langle m+k|\psi\rangle$ and with constant coefficients.

In the light of the analysis of a few similar examples [4–6], let us formulate now a general approach to the Schrödinger bound state problem represented by eq. (5) and by its extrapolation (8).

In the first step, we shall consider eq. (8) as a difference equation with some complete set of general solutions

$$\psi_m^{(i)}, \quad i=1, 2, \ldots, N_0,$$

$$\psi_m^{(-j)}, \quad j=1, 2, \ldots, N_1, \tag{9}$$

which are "asymptotically physical" and "asymptotically unphysical", respectively. The examples of ref. [4] with

$$\psi_m^{(i)} \sim \gamma_i^m, \quad \|\gamma_i\| < 1, \quad i=1, 2, \ldots, N_0,$$

$$\psi_m^{(-j)} \sim \delta_j^m, \quad \|\delta_j\| > 1, \quad j=1, 2, \ldots, N_1, \tag{10}$$

may be recalled as an illustration and exhibit clearly their respective asymptotic decrease (normalizability) and increase (divergence) in the simplest form.

On the basis of a close analogy between difference and differential equations [4,7], we may write now a general asymptotically physical solution of eq. (8) as a superposition of the "dominated" [6] $i>0$ components (9),

$$\psi_m^{(\mathrm{Jost})} = \sum_{i=1}^{N_0} \psi_m^{(i)} c_i . \tag{11}$$

Similarly, a final physical solution may be obtained by an imposition of the corresponding "boundary conditions in the origin" [4]. In the present case, they are represented by the first N_0 rows of eq. (8),

$$\sum_{j=1}^{N_0} A_{ij} c_j = 0, \quad i=1, 2, \ldots, N_0,$$

$$A_{ij} = \sum_{n=0}^{M} (\langle i-1|H|n\rangle - E\delta_{i-1,n})\psi_n^{(j)} . \tag{12}$$

In particular, their (generalized [8]) eigenvalue-problem form specifies the energies as the roots of the equation

$$\det A = 0$$

and determines also the physical values of the coefficients c_i in (10).

In the spirit of ref. [5], the matrix A may easily be determined numerically. Indeed, we may start from any initial (and not necessarily asymptotically physical) state $|\psi^{(\mathrm{trial})}\rangle$ compatible only with our eq. (8) at some very large $k=K\gg 1$,

$$\psi_m = \psi_m^{[0]} = \langle K+m|\psi^{(\mathrm{trial})}\rangle, \quad m=0, 1, \ldots, M .$$

Then, step-by-step, we may also employ our eq. (8) (with the decreasing k's) as a "selfconsistent" definition of some new sets of vectors $\psi_m^{[1]}, \psi_m^{[2]}, \ldots$, such that

$$\psi_m^{[l+1]} = \psi_{m-1}^{[l]} \quad (\doteqdot \langle K+m-l-1|\psi^{(\mathrm{trial})}\rangle),$$

$$l=0, 1, \ldots . \tag{13}$$

Obviously, (e.g., due to the relations of type (10)), the unphysical components of $\psi_m^{[L]}$ become strongly suppressed for $L\gg 1$ by the iterations (13). Numerically, we may therefore define the normalizable $i>0$ components (9) of $\psi_m^{(\mathrm{Jost})}$ as the $l=L\gg 1$ iterates (13) initialized by some N_0 independent vectors $|\psi^{(i\text{th trial})}\rangle$, in complete analogy with the examples of refs. [5] and [6].

For an illustration of the efficiency of the present iterative algorithm, we have chosen the $N_0=2$ anharmonic oscillator hamiltonian

$$H = -\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + \mu r^2 + \nu r^4 , \tag{14}$$

Table 1
Precision of energies $E^{\text{computed}} - E^{\text{exact}}$ for the anharmonic oscillator (0.23_{-5} denotes 0.23×10^{-5}).

Dimension N	Ground state		Excited state	
	variational method	present method	variational method	present method
10	0.23_{-5}	-0.18_{-5}	0.30_{-3}	-0.71_{-2}
11	0.80_{-6}	-0.39_{-6}	0.43_{-4}	-0.39_{-2}
12	0.18_{-6}	-0.32_{-5}	0.73_{-5}	-0.12_{-2}
13	0.23_{-7}	-0.20_{-5}	0.70_{-5}	-0.22_{-3}
14	0.55_{-8}	-0.64_{-6}	0.58_{-5}	-0.18_{-5}
15	0.55_{-8}	-0.10_{-6}	0.26_{-5}	$+0.98_{-6}$
16	0.37_{-8}	$+0.25_{-8}$	0.85_{-6}	-0.98_{-5}
17	0.16_{-8}	-0.44_{-9}	0.15_{-6}	-0.86_{-5}
18	0.49_{-9}	-0.57_{-8}	0.27_{-7}	-0.40_{-5}
19	0.96_{-10}	-0.53_{-8}	0.12_{-7}	-0.12_{-5}
20	0.15_{-10}	-0.23_{-8}	0.12_{-7}	-0.16_{-6}

with $\mu=\nu=l+1=1$. In the usual (harmonic-oscillator) pentadiagonal matrix representation, we may consider eq. (3) (input) with $M=N+2$. For the ground state ($E=1.392...$) and the first excited state ($E=8.655...$) with $L=50$ iterations, $N_0=2$ and K=L, our "oblong-truncation" energies are listed in table 1. They seem to form the very precise "anti-variational" lower estimates, complementing the standard variational upper bound of the exact energies. This property represents a methodically promising phenomenon to be analyzed in more detail in the future.

Summarizing our considerations in a partitioned matrix notation, we are inserting the ansatz $\psi_m^{(\text{Jost})}$ with

$$\psi_m^{(i)} = U_{m+1,i}, \quad m=0, 1, ..., N, \qquad \psi_{N+j}^{(i)} = W_{ji}, \quad i,j=1,2,...,N_0\,, \tag{15}$$

in our Schrödinger equation (5), and obtain

$$(BU+DW)\begin{pmatrix} c_1 \\ c_2 \\ \cdots \\ c_{N_0} \end{pmatrix}=0\,, \tag{16}$$

where B and D denote the corresponding submatrices of $H-EI$. The product

$$\begin{pmatrix} U \\ W \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \\ \cdots \\ c_{N_0} \end{pmatrix}=\tilde{\psi}$$

coincides with the first $M+1$ projections of the wavefunction $|\psi\rangle$. In their more traditional square-matrix definition [9]

$$\begin{pmatrix} B & D \\ D^+ & F \end{pmatrix}\tilde{\psi}=0 \tag{17}$$

the last N_0 rows become identities whenever $D^+U+FW=0$.

Thus, our iterative procedure completed by the formula

$$F=-D^+UW^{-1} \tag{18}$$

may also be interpreted as a new nonperturbative construction of the Feshbach effective hamiltonian [9].

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